EXP NO:3.a Conduct an experiment to show data visualization using line plot

AIM:
To conduct an experiment to show data visualization using a line plot.

PROGRAM:

python

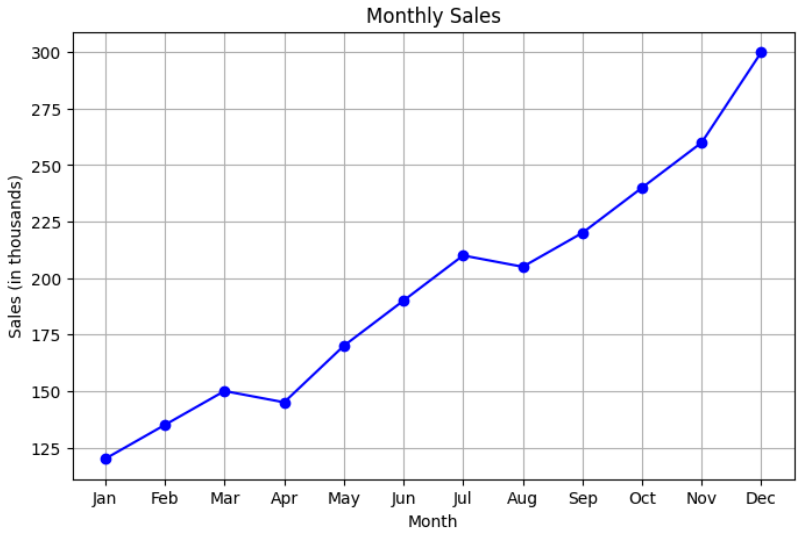
```
import pandas as pd
import matplotlib.pyplot as plt

# Monthly sales data
data = {
    "Month": ["Jan", "Feb", "Mar", "Apr", "May", "Jun",
              "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"],
    "Sales": [120, 135, 150, 145, 170, 190, 210, 205, 220, 240, 260, 300]
}

df = pd.DataFrame(data)
print(df)

# Line plot
plt.figure(figsize=(8, 5))
plt.plot(df["Month"], df["Sales"], marker='o', color='blue', linestyle='-')
plt.title("Monthly Sales")
plt.xlabel("Month")
plt.ylabel("Sales (in thousands)")
plt.grid(True)
plt.show()
```

OUTPUT:



RESULT:
Thus the program to show data visualization using a line plot was executed successfully.

EXP NO:3.b Conduct an experiment to show data visualization using bar chart

AIM:
To conduct an experiment to show data visualization using a bar chart.

PROGRAM:

python

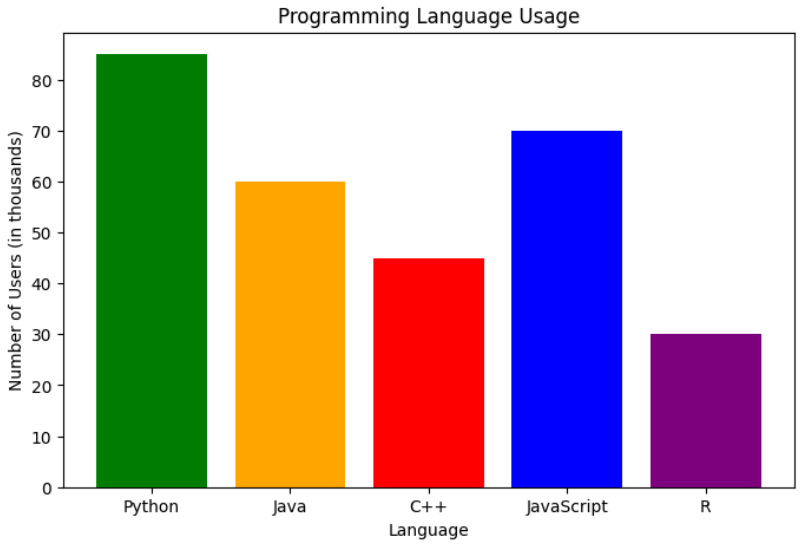
```
import pandas as pd
import matplotlib.pyplot as plt

# Programming language popularity data
data = {
    "Language": ["Python", "Java", "C++", "JavaScript", "R"],
    "Users": [85, 60, 45, 70, 30]
}
```

```
df = pd.DataFrame(data)
print(df)

# Bar chart
plt.figure(figsize=(8, 5))
plt.bar(df["Language"], df["Users"], color=['green', 'orange', 'red', 'blue', 'purple'])
plt.title("Programming Language Usage")
plt.xlabel("Language")
plt.ylabel("Number of Users (in thousands)")
plt.show()
```

OUTPUT:



RESULT:
Thus the program to show data visualization using a bar chart was executed successfully.

EXP NO:3.c Conduct an experiment to show data visualization using pie chart

AIM:
To conduct an experiment to show data visualization using a pie chart.

PROGRAM:

python

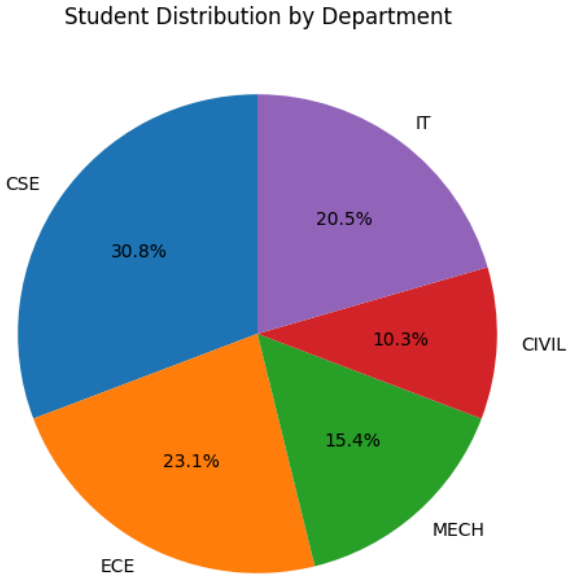
```
import pandas as pd
import matplotlib.pyplot as plt

# Student department distribution data
data = {
    "Department": ["CSE", "ECE", "MECH", "CIVIL", "IT"],
    "Students": [120, 90, 60, 40, 80]
}

df = pd.DataFrame(data)
print(df)

# Pie chart
plt.figure(figsize=(6, 6))
plt.pie(
    df["Students"],
    labels=df["Department"],
    autopct='%1.1f%%',
    startangle=90
)
plt.title("Student Distribution by Department")
plt.show()
```

OUTPUT:



RESULT:

Thus the program to show data visualization using a pie chart was executed successfully.